Decoder decay and its correction over years in intracortical brain–computer interfaces

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Abstract

Implanted brain–computer interfaces let people with paralysis move cursors and communicate, but their decoders lose accuracy as recordings change. We measured this decay with one pipeline in public recordings from three monkeys and six people, and electrode changes in 14 people. Decay differed widely between individuals: a fixed decoder lost half its accuracy within a day in one participant and only after 206 days in another, and weak regularization exaggerated decay. In people, about 100 labeled trials restored most accuracy. The label-free alignments we tested added nothing beyond normalizing each channel per session, although an alignment fitted with labels showed that better alignments existed. Even that ceiling left 15–58% of the loss, and in a pilot a pretrained foundation decoder also needed labels. Most brief electrode silences looked like noise; month-long silences were real but often reversed. We release a 1,824-pair benchmark with gaps up to 480 days.

Keywords: brain–computer interface, intracortical recording, Utah array, neural drift, decoder recalibration, electrode stability

Introduction

Intracortical brain–computer interfaces (BCIs) record from small populations of neurons and turn their activity into commands. They have allowed people with paralysis

to move robotic arms and computer cursors[1–4], to type by attempted handwriting[5] and to speak through a decoded voice[6–8]. Long-term use over years has been described[9,10]. For these devices to become everyday tools, their decoders must stay accurate with little effort from the user.

The recorded signals are not stable. Within a day, signals shift as electrodes move or neurons change their firing[11]. Across days, the activity seen by each electrode changes[12–15], and over months and years, arrays gradually lose the ability to record spikes[16–19]. Decoders lose accuracy over time[13,20–22], and most systems recalibrate often, which takes time away from the user[23–25]. Many methods aim to reduce this cost. Decoders can be trained on data from many days[26], updated during use[22,24,27,28], or given input layers learned for each day[5,29]. Neural activity can be aligned to a reference day without new labels[30–34], and monitoring scores have been proposed to signal when recalibration is needed[35].

How fast do decoders decay in different individuals, and how much of that decay comes from the decoder itself? Which corrections recover the loss, and with how much labeled data? What happens at the electrodes over the years? Answers have come mostly from single datasets, spans of weeks and different baselines[36], so results are hard to compare. In one participant, for example, a decoder whose channel means were updated still failed after months, whereas adjusting its weights recovered it[22]. Whether such results hold across individuals and species is not known.

Public long-term recordings now make a common analysis possible[19,21,31,35,37]. We analyzed four such datasets with one pipeline. We trained decoders on one day, tested them on later days and applied a series of corrections, from none to full retraining. Figure 1 shows the problem in one participant: over seven years, one array lost about half of its electrodes with spikes, the remaining spikes shrank by about half, and a decoder trained 478 days earlier failed until each channel was re-weighted. We asked how fast decoders decay in each individual, which corrections recover the loss and with how many labels, why label-free alignment falls short, and what changes at the electrodes. We release the code, a benchmark for decoder transfer over years, a calibrated simulator and all derived results.

Results

Datasets and approach

We assembled four public datasets (Fig. 2a and Supplementary Table 1). Decoder analyses used nine individuals. Monkey N was recorded over 1,242 days with Utah arrays in motor cortex during a two-finger task[21,38,39]. Monkeys C and M were recorded over three and 1.5 years during reaching[31,37]. BrainGate participants T6, T5, T9, T7, T10 and T8 contributed 35–124 closed-loop cursor sessions each, over 0.9–7.2 years (483 sessions in total; Supplementary Table 14)[19]. Three further participants were excluded by a rule fixed when the first of them failed (Methods): the decoders of T2 and T3 did not work even on the day they were trained, and T11's recordings jumped in scale between recording blocks (Supplementary Note 4). Electrode analyses used

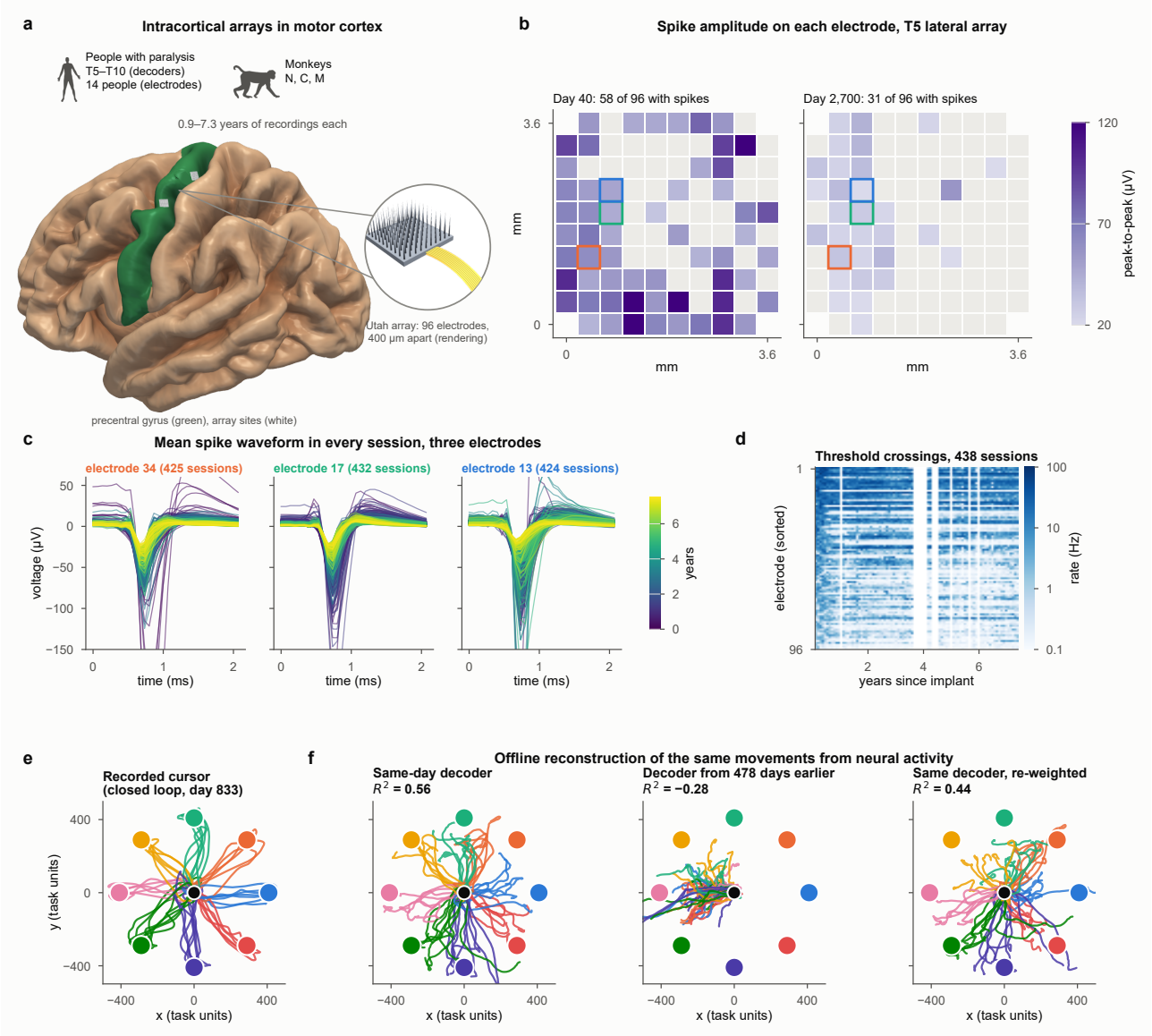


Fig. 1 Recordings change over years, and decoders decay with them. a, Approximate array sites (white) on the precentral gyrus (green) of a standard cortical surface, and a rendered Utah array. **b**, Spike amplitude on each electrode of T5's lateral array, 7.3 years apart. Grey cells had no spikes; outlines mark the electrodes in **c**. **c**, Mean spike waveforms of three electrodes in every session, coloured by time since implant. **d**, Threshold-crossing rate of every electrode on the same array, in 30-day bins. **e**, Cursor paths in a closed-loop session of T5. Colours mark targets. **f**, The same movements reconstructed offline by three decoders. The decoder trained on day 355, 478 days earlier, fails until each channel is re-weighted (Methods).

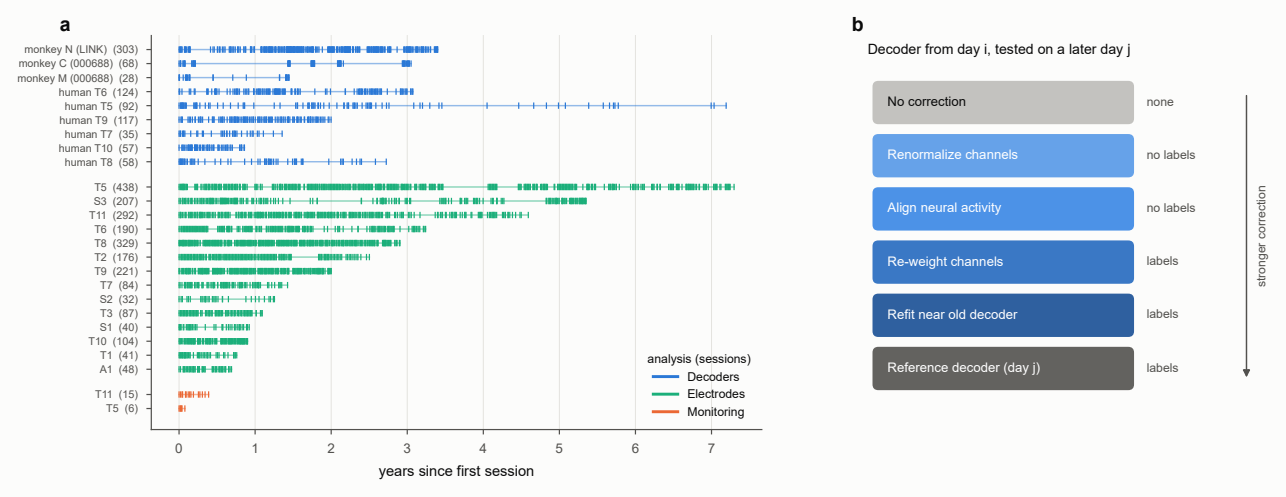


Fig. 2 Datasets and the correction ladder. a, Sessions analyzed, by individual. Each tick is one session. Numbers give sessions per analysis. **b**, The correction ladder. A decoder trained on day i is tested on a later day j after increasingly strong corrections. A decoder trained on day j is the reference.

all 20 arrays from 14 BrainGate participants (2,289 sessions) and monkey N. Monitoring analyses used closed-loop sessions in which participants T11 and T5 controlled a cursor with a fixed decoder[35].

For each individual, we trained a linear decoder on one session and tested it on later sessions at gaps of 1 to 480 days. Before testing, every channel of the later session was z-scored with that session's own mean and standard deviation. We call this step renormalization. It needs no labels, and the result is the renormalized fixed decoder. We then applied a ladder of corrections (Fig. 2b and Methods). Label-free corrections updated channel means or aligned neural activity to the training day. Corrections with labeled trials from the later day either re-weighted each channel in front of the otherwise unchanged decoder, or refitted the decoder while shrinking it toward the old one. Renormalization (no labels) and re-weighting (labels) both act on single channels, and we keep the two names distinct throughout. A decoder trained on the later day served as the reference. We report retention, the ratio of a decoder's accuracy (coefficient of determination, R^2) to that of the reference on the same test trials.

Decay rates differ widely between individuals

Without any correction, decoders failed: reusing the training day's normalization gave median R^2 below zero at most gaps (Supplementary Table 2). We use the renormalized fixed decoder as the baseline.

A single night already cost accuracy. After exactly one day, retention was 0.70–0.78 in the monkeys and 0.56–0.58 in the 4 participants with at least three one-day pairs (Fig. 3b; intervals in Supplementary Table 17).

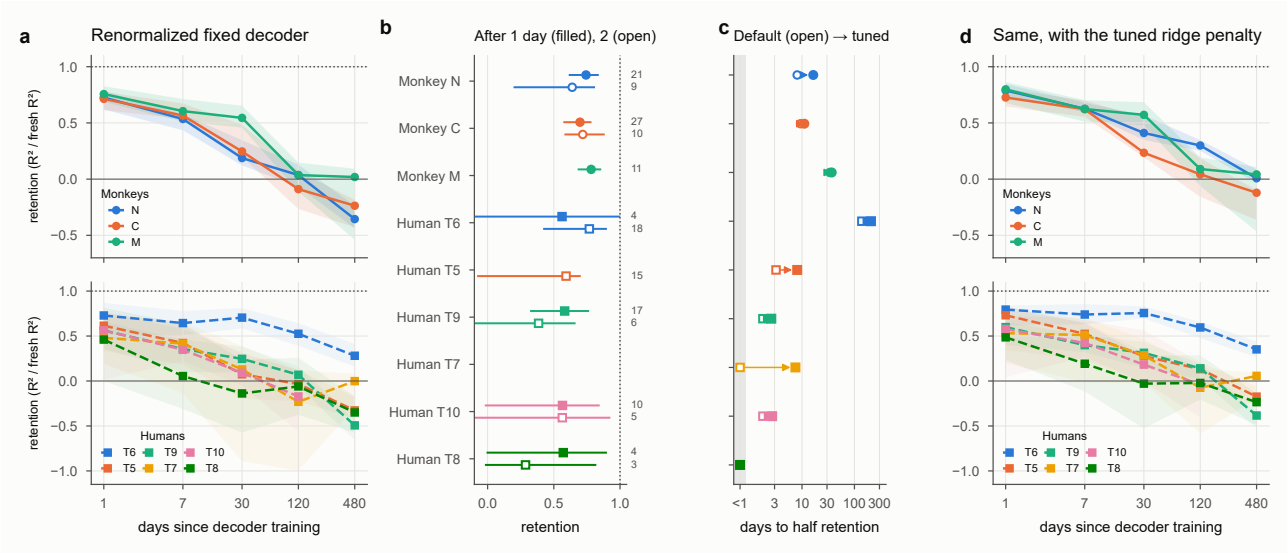


Fig. 3 Decoders decay at rates that differ between individuals. a, Retention of the renormalized fixed decoder, the R^2 of the old decoder divided by that of the reference decoder, against days since training, for monkeys (top, circles) and humans (bottom, squares). Bands show 95% confidence intervals. **b**, Retention after exactly one day (filled) and two days (open). Numbers give session pairs. **c**, Days until retention falls to one half, with the default ridge penalty (open) and the tuned penalty (filled). Values in the shaded band (<1) were already below one half at the first tested gap, which pools one- and two-day pairs. **d**, As in **a**, with the tuned penalty.

Longer-term decay differed far more. Retention fell to one half in under a day in T7 and T8, within 3.2 days in T5, T9 and T10, within 8.1–34 days in the monkeys and after 138 days in T6 (Fig. 3a,c). Values under one day are censored and the intervals are wide (Supplementary Table 17), so we compare individuals by range, not by ratio. With the better-regularized decoder of the next section, the range stayed wide, from under one day in T8 to 206 days in T6. A mixed model of retention against log elapsed time, with random effects for training and test sessions, confirmed that the decay curves differed between individuals ($P = 1.0 \times 10^{-28}$; participants alone, $P = 2.8 \times 10^{-20}$; Methods). This held for every outcome definition we tried, and a rank-based test with one value per training session, which does not assume equal variances, agreed at gaps of 7 days or more ($P \leq 2.1 \times 10^{-5}$ in the participants; Supplementary Note 4). After a single night the participants did not differ detectably ($P = 0.19$). In monkey N and five participants, the reference decoder did not decline from short to long gaps (Supplementary Table 2), so the old decoder no longer fits the new day even though the recordings decode as well as before. In T10 the reference fell slightly (median R^2 0.34 after one day, 0.27 after 120 days), and monkey M fluctuated without a trend. In monkey C the reference decoder itself fell at the longest gaps (median R^2 0.18 and 0.15 after 120 and 480 days, against 0.27–0.45 at gaps up to 30 days), so there the recordings also became harder to decode. Retention divides this out.

Weak regularization exaggerates decay

Part of the measured decay came from the decoder rather than the recordings. We had adopted the ridge penalty used in the original monkey N study[21] ($\alpha = 0.1$; $\alpha = 1$ for the sparser spike counts of monkeys C and M). On 768 to 1,536 lagged features, this is almost no regularization. A stronger penalty can be chosen from training-day data alone: we took the largest penalty whose same-day accuracy stayed within 1% of the best (the 1% rule). The rule chose $\alpha = 10^4$ in monkeys N and C and in T6 and T5, 3×10^4 in the other four participants and 10^3 in monkey M (Supplementary Table 9). We first saw the benefit of stronger penalties in monkey N's cross-day curves. The rule, which uses same-day data only, then chose 10^4 in the first individuals we analyzed (monkeys N and C and T6), and we fixed this value for everyone in all later analyses.

With $\alpha = 10^4$, the time to half retention about doubled or more in monkey N (8.1 to 16 days), T5 (3.2 to 8.1 days) and T7 (under one day to 7.5 days). It grew by about half in T6, T9 and T10 and changed little in monkeys C and M and in T8 (Fig. 3c,d). Absolute R^2 on later days rose in all nine individuals, by medians over training sessions of 0.003 to 0.061. The 1% rule did not lower same-day accuracy (Supplementary Table 9), and ordinary same-day cross-validation of the penalty gave almost the same cross-day accuracy (Methods). A special rule is therefore not needed. What matters is that decay is compared only between decoders whose penalty was tuned at all. We use $\alpha = 10^4$ for the remaining decoder analyses unless stated.

Label-free alignment misses alignments that labels can find

Several methods realign neural activity to a reference day without labels[30–34]. Their benefits were usually measured against a fixed decoder or against one whose channel means alone were updated[22,33]. We compared such corrections with renormalization (Fig. 4a; differences and P values in Supplementary Table 3).

Updating channel means alone was worse than renormalization in every individual. Adding an orthogonal Procrustes rotation of the dominant neural subspace[40] lowered R^2 slightly in every individual (median changes -0.063 to -0.004). Matching the full channel covariance (CORAL[41]) was no better in the monkeys and worse in all six participants. Our implementation of the factor-analysis stabilizer[30] was worse in seven of the nine individuals, did not differ in T7 and was nominally better in T5, which was not significant after Holm correction. That method was designed for closed-loop use with a latent decoder. Adversarial alignment[32–34] was not tested. Comparisons against mean-only baselines[22] should also report renormalization.

Why did alignment fail? Either no good alignment existed, or it could not be found without labels. To tell these apart, we fitted the best linear alignment of the dominant 16-dimensional subspace with labels: a free map that can rotate, scale and mix activity within the subspace, fitted on all labeled training trials of the later day and scored on its held-out test trials (Methods). It estimates how much a linear alignment of this subspace can recover. At gaps of 120 and 480 days, this ceiling recovered 48–85% of the loss in the monkeys and 42–77% in the participants, whereas the label-free rotation recovered -27 to -10% and -18 to 0% (Supplementary Table 18). Good alignments

therefore existed, but the label-free method did not find them. Even the ceiling left 15–58% of the loss, a part that no linear alignment of this subspace recovered.

The ceiling also separated the two species. In all three monkeys, it recovered more than re-weighting each channel on the same trials (15–46%; Holm-corrected $P < 0.05$ in 3 of 3). Their decoders predicted four signals and could not be rescued when their channels were scrambled (see below), so in the monkeys a large share of the change was a remixing of activity across the population. In all six participants, re-weighting recovered more than the ceiling (75–86%; 6 of 6 after Holm correction), although only slightly in T6. With a two-dimensional target, per-channel weights are flexible enough to rescue even scrambled decoders, so this result fits per-channel change in people but does not prove it.

Neural divergence adds little to elapsed time as a monitor

A monitor that warns when a decoder is failing could make recalibration less frequent, and neural activity can even flag single decoding errors as they happen [42]. MINDFUL scores instability as the divergence between distributions of neural features, or of decoder outputs, on a test day and a reference day [35]. Such scores correlate with decoder error, but both grow with time since calibration.

Offline in monkey N (537 session pairs), elapsed time alone correlated with decoder accuracy (Spearman $|\rho| = 0.67$; Fig. 4b). Neural divergence correlated more weakly ($|\rho| = 0.34$), and its partial correlation fell to 0.16 after controlling for elapsed time; output divergence behaved the same way. Adding both divergences to elapsed time slightly increased the cross-validated error of predicted R^2 (0.090 versus 0.085). A pilot on the first year of monkey N tested a broader set of label-free statistics, and none improved on elapsed time (Supplementary Note 1).

We then re-analyzed the public MINDFUL data, in which T11 and T5 controlled a cursor in closed loop with a fixed decoder (Fig. 4c and Supplementary Table 4). In T11 (15 days), neural divergence correlated with angular error across 60-s windows ($r = 0.82$), consistent with the original report [35]. Most of this association was shared with elapsed time (partial $r = 0.37$), and adding neural divergence did not improve leave-one-day-out prediction of error (17.5° versus 17.1°). Output divergence kept a partial correlation of 0.73, and adding it reduced the error to 11.9°. So output divergence carried information that elapsed time did not. Part of this link may be mechanical, because output divergence and angular error are both computed from the cursor's motion, and the window-level statistics are nested within few days. The six days available for T5 allow only a descriptive comparison. Within these limits, neural divergence was no better a guide than elapsed time, whereas output divergence deserves a test on more days.

A few minutes of labeled data restore decoders in people

The tuned penalty also mattered for recalibration: the 1% rule raised cross-day R^2 at every gap without lowering same-day accuracy (Fig. 5a,b). How much labeled data does recalibration need? In the six participants, retention pooled over all gaps rose from 0.23 without labels to 0.67 with 50 labeled trials and 0.72 with 100 for a freshly

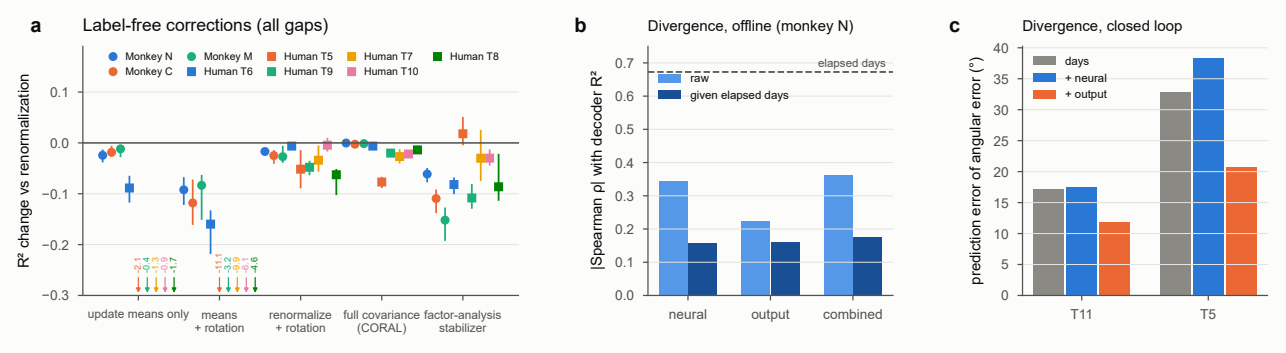


Fig. 4 Label-free corrections and neural monitoring add little. a, Change in R^2 for five label-free corrections, relative to renormalization. Points show medians over training sessions; bars show 95% confidence intervals. Values below the axis are printed. **b**, Offline monitoring in monkey N. Bars show how strongly each divergence score correlates with decoder accuracy, before (light) and after (dark) removing the effect of elapsed time. The dashed line marks elapsed time alone. **c**, Closed-loop monitoring in the public MINDFUL data. Bars show the error in predicting cursor angular error from elapsed days alone, or with a divergence score added.

trained, strongly regularized decoder (Fig. 5c and Supplementary Table 10). Refitting while shrinking toward the old decoder reached 0.56 and 0.66, and re-weighting reached 0.48 and 0.57. Movement periods lasted 1.6–3.9 s in these sessions, so 100 trials take a few minutes. Recovery was slowest in T8, where 100 trials restored a retrained decoder to 0.46, against 0.74–0.92 in the other participants (Supplementary Table 10). With 10 or 20 trials, refitting toward the old decoder was better (0.42 against 0.25 with 10 trials). From 50 trials on, a fresh decoder learned more from the same trials, so in people the old decoder helped only when very few labels were available. Re-weighting indicates what changes, but it is not an efficient way to recalibrate in these sessions.

With very few trials, refitting can make a decoder worse. In monkey N (86 session pairs), choosing the shrinkage strength by cross-validation on 10 trials made 10.5% of recalibrations worse than no recalibration, and a fixed strong shrinkage made 1.2% worse (Fig. 5d). This difference was not significant after correction for the five data budgets (Supplementary Table 10).

A pretrained foundation decoder still needed labels in a pilot

Large decoders pretrained on many recordings might be expected to resist drift. We tested NDT3[43], a transformer pretrained on about 2,000 hours of intracortical motor recordings, in monkey N and in participants T5 and T8, none of which were in its pretraining data (Methods). NDT3 and the ridge decoder received the same threshold crossings, on 12 session pairs per individual. Within a day, NDT3 was about twice as accurate as the ridge decoder in monkey N (median R^2 0.42 versus 0.20) but less accurate in T5 (0.34 versus 0.45) and T8 (0.22 versus 0.28). Without an update, both decoders fell below $R^2 = 0$ on later days in all three individuals. Continuing NDT3's own self-supervised training on the new day, without labels, left its median retention below zero in all three (−0.43, −0.16 and −0.08). With 300 labeled trials, NDT3

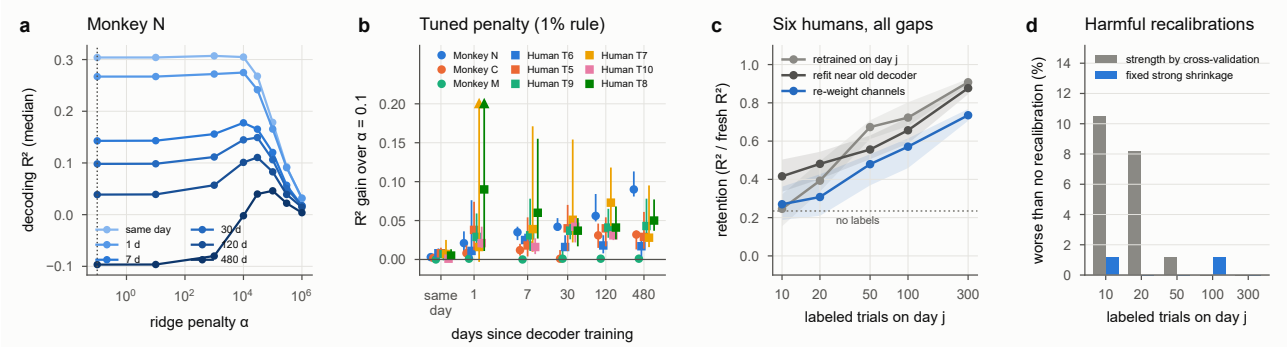


Fig. 5 Simple recipes for recalibration. a, Decoding R^2 in monkey N against the ridge penalty, on the training day (lightest) and on later days (darker). The dotted line marks $\alpha = 0.1$. **b**, Gain from the 1% rule over $\alpha = 0.1$, the smallest penalty tested, for every individual (including monkeys C and M, whose default was $\alpha = 1$), per individual and gap (medians over training sessions with 95% confidence intervals; intervals beyond the axis are clipped, arrows). **c**, Retention in the six human participants against the number of labeled trials (pairs whose later day had at least 300 labeled training trials), for a retrained decoder, a refit shrunk toward the old decoder and re-weighting. The dotted line marks the renormalized fixed decoder. **d**, Recalibrations in monkey N that made the decoder worse than no recalibration, with the shrinkage chosen by cross-validation (grey) or fixed (blue).

recovered fully in monkey N (median R^2 0.41 versus 0.18 for the ridge decoder refit toward its previous weights), but it stayed below the refit ridge decoder in T5 (0.26 versus 0.41) and T8 (−0.04 versus 0.18; Supplementary Table 16 and Supplementary Note 5). This was a pilot, with three training sessions per individual and two learning rates. NDT3 was weaker than the ridge decoder in people even within a day, so better fine-tuning may close part of that gap. Within this pilot, a large pretrained decoder did not remove the need for labeled recalibration.

Channel tuning is partly preserved over months

Label-free corrections left much of the loss in place, and labels were needed to recover it. To see what kind of change remained, we re-weighted each channel in front of the otherwise unchanged decoder, using labeled trials from the later day. Re-weighting lets each channel's contribution grow, shrink or vanish, but keeps the decoder's pattern across time lags and outputs. In the participants it recovered 12–84% of the loss at gaps up to 480 days, least in T8 and most in T5 and T9 (Fig. 6a,d). In monkeys N, C and M it recovered −5–21%, 22–38% and 19–35%.

Scrambling the channels tests whether re-weighting repairs the old decoder or simply builds a new one. In the participants, re-weighting rescued even scrambled decoders (retention 0.33–0.77, against 0.52–0.89 with correct channels; Fig. 6b), because a two-dimensional target can be fitted from almost any set of weights. In the monkeys, whose decoders predicted four kinematic signals, scrambled decoders could not be rescued (retention −0.19, 0.02 and 0.01). High recovery in people is therefore not, by itself, evidence that each channel kept its role. Decoding a two-dimensional movement direction in monkeys C and M raised the share recovered to 36–60% and

30–48% (Supplementary Table 7), but we did not test scrambled decoders in that setting.

When the weights could not change sign, correct channels beat scrambled ones in every individual (in the participants, 0.38–0.89 against 0.10–0.57), and the constraint cost at most 0.14 in retention with correct channels (T8). In the monkeys, drift therefore preserves part of each channel's tuning pattern, even though much of the change was a remixing across channels. In the participants the evidence is weaker, because the sign constraint penalizes flipped channels by design.

Re-weighting also tracked measured changes at each electrode (Fig. 6c and Supplementary Table 5). We compared the fitted weights with each electrode's change in threshold-crossing rate between the two days, taken from the BrainGate yield data. This rate is a different feature from the decoder's spike-band power, although both come from the same electrode signal. Channels whose firing rose got larger weights in 6 of 6 participants (6 after Holm correction), and channels whose impedance rose got smaller weights in T9, T7, T10 and T8 (T9, T10 and T8 after Holm correction). The effects are weak for single channels (median Spearman ρ up to 0.11) but absent after shuffling channels.

A recurrent network decoder gave the same picture. Re-weighting its inputs recovered 27–101% of its loss in the participants, overlapping the linear decoder, and more than for the linear decoder in the monkeys (Fig. 6d and Supplementary Table 6). Angles between decoders fitted on different days shrank when one scale per channel was allowed, but much of that came from the added freedom (Supplementary Table 8).

Brief electrode silences look like fluctuation

How do single electrodes change over years? The BrainGate yield data report threshold-crossing rates and impedance for every electrode in every session[19], and monkey N provides matching data. An electrode was active when its threshold-crossing rate exceeded 2 Hz, the definition used by BrainGate[19]. Impedance declined over years in both species, whereas array yield changed in both directions (Fig. 7a,b). Yield fell between the first and last three sessions in 10 of 20 human arrays (Supplementary Table 11). Impedance fell in 17 of 18 human arrays with measurements (median Spearman ρ with time -0.92). In monkey N it fell from a median of 302 kΩ in the first year to 171 kΩ in the fourth ($\rho = -0.88$). Both trends replicate earlier reports[16–19,44]. Spikes also became smaller. On T5's lateral array, 62 electrodes had spikes both in the first 200 days and after day 2,000. Their median spike amplitude fell from 55 to 29 μV (median ratio 0.51, interquartile range 0.44–0.59), and it was smaller on 61 of them ($P = 8.4 \times 10^{-12}$, Wilcoxon signed-rank test; Fig. 1b,c).

Electrodes often fell silent and later recovered[14,16], but most such events could not be told apart from fluctuation (Fig. 7c and Supplementary Table 11). Of 730 initially active human electrodes that stayed below threshold for at least three consecutive sessions, 80% later recovered for at least three sessions. When each electrode's session order was shuffled 200 times, which destroys any temporal process, the recovery rate was 78% (95% range 75–81%; permutation $P = 0.075$). Recovery was also not detectably above the shuffled rate when silence required rates below 0.5 Hz, or with a fixed threshold in microvolts for each electrode, which prevents changes in noise level

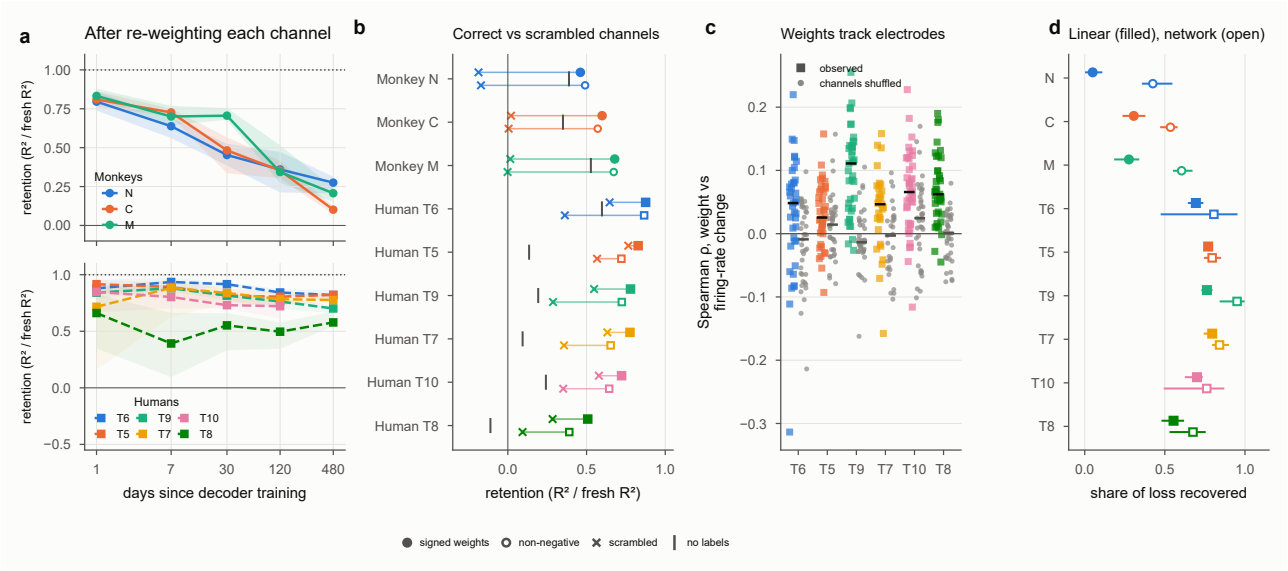


Fig. 6 Channel tuning is partly preserved over months. a, Retention after re-weighting each channel in front of the unchanged decoder, for monkeys (top) and humans (bottom). **b**, Retention after re-weighting the correct decoder (circles and squares) or a decoder with scrambled channels (crosses). Upper rows use signed weights; lower rows use non-negative weights. Ticks mark the renormalized fixed decoder. **c**, Correlation between the fitted channel weights and each electrode's change in firing rate. Each point is one training session. Grey points show the same after shuffling channels. **d**, Share of the lost accuracy recovered by re-weighting, for linear (filled) and recurrent (open) decoders. Bars show 95% confidence intervals.

from moving the threshold. A difference that is not detected is not proof of equivalence, but these silences give no sign of a process beyond session-to-session fluctuation. Silences lasting at least 30 days were different. They were twice as common as in the shuffled data (168 versus 79 on average), and 57% of them ended in recovery, against 44% in the shuffled data (95% range 33–54%; $P = 0.010$). Recovery here means that the electrode again recorded spiking, not necessarily from the same neuron.

Spiking declined slightly faster at the edge of the array than in its interior, in 13 of 17 human arrays (one-sided $P = 0.036$, with the array as the unit), but the effect weakened after adjusting for each electrode's initial firing rate (one-sided $P = 0.087$; Fig. 7d). Models in which micromotion strains tissue most at the edges predict such a pattern[45], but our evidence for it is weak.

A benchmark and a simulator for testing new corrections

Methods for stable decoding are hard to compare without a common test bed[36], and existing benchmarks span days to months. We released our session pairs as a benchmark for decoder transfer over years (Methods). It fixes 1,824 pairs from the nine individuals at gaps of 1 to 480 days (1,684 with a working reference decoder), defines which data a method may use on the new day, and scores retention with confidence intervals over training sessions. The correction ladder serves as its baseline.

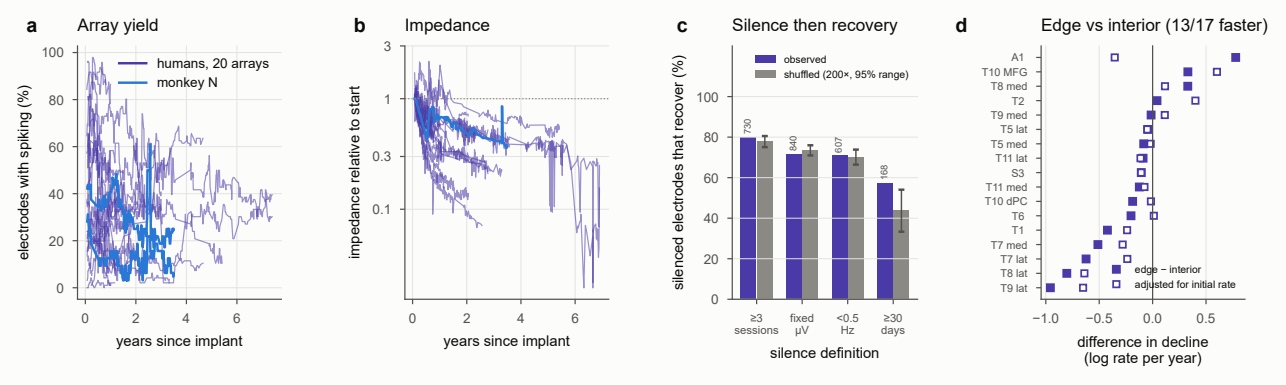


Fig. 7 Brief electrode silences look like fluctuation. a, Percentage of electrodes with spiking activity over years, for 20 human arrays (purple) and two arrays in monkey N (blue). **b**, Median impedance of each array relative to its first sessions. **c**, Percentage of silenced human electrodes that later recovered, observed (purple) and after shuffling session order 200 times (grey, with 95% range). Four definitions of silence are shown. Numbers give silenced electrodes. **d**, Difference in the yearly decline of firing rate between edge and interior electrodes, per human array, before (filled) and after (open) adjusting for initial firing rate.

Across individuals, the renormalized fixed decoder kept a median retention of 0.73 after one day and −0.06 after 480 days. With 300 labeled trials, re-weighting each channel kept 0.83 and 0.64, and a refit shrunk toward the old decoder kept 0.97 and 0.98 (Supplementary Table 15). With 300 labeled trials, a retrained decoder in many human sessions uses all training trials and so equals the reference; budgets of 10 to 100 trials are the informative ones. The evaluation labels are public, so there is no hidden test set, and the valid-pair rule depends on the reference decoder's test accuracy; the manifests fix the valid pairs so that every method is scored on the same ones.

We also built a feature-level simulator that takes a real session and returns it as it might appear days to years later (Methods). It combines mixing of signals between channels, gradual turnover of each channel's tuning and switching of electrodes between active and silent states. Switching rates came from monkey N's electrode statistics. The mixing and turnover parameters were calibrated so that the simulated correction ladder matched monkey N's during its first 700 days. The fit had a root-mean-square error of 0.07 in retention (Fig. 8a).

We tested the simulator on the later days, which it had not seen (Fig. 8b). It beat a simulator without drift everywhere. It beat a simpler baseline, the median retention measured in the first 700 days, only for corrections it was not fitted to (median absolute error 0.06 versus 0.16); for the fitted corrections and for different amounts of labeled data, that empirical curve was as good or better. The simulator is therefore most useful for asking how a new correction might behave, not for predicting known ones. A calibration on the later years gave weaker mixing and slower turnover (time constant 527 versus 360 days; Fig. 8c and Supplementary Table 12), so drift may slow as an implant ages, at least in this monkey. Training decoders on simulated futures helped little (Supplementary Table 13).

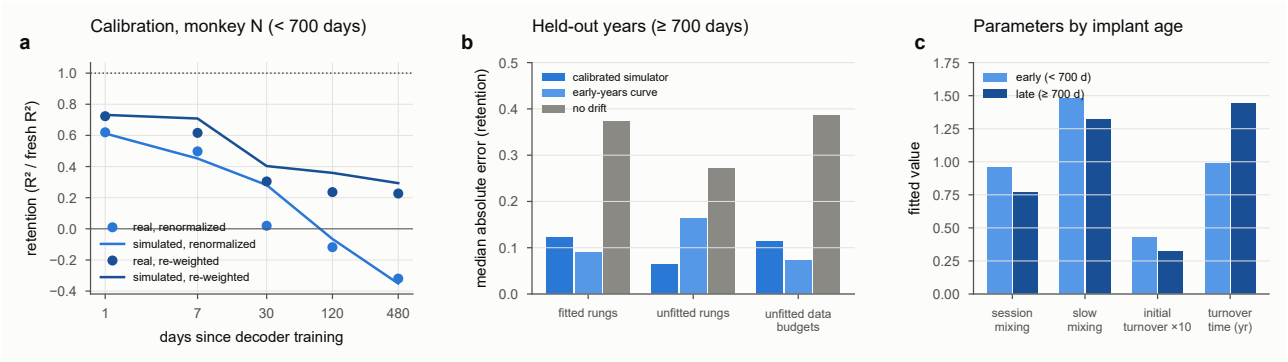


Fig. 8 A calibrated simulator for testing new corrections. **a**, Calibration on monkey N sessions before day 700. Points show real retention; lines show the simulator. **b**, Prediction error on sessions after day 700, for the simulator (dark blue), the median retention measured before day 700 (light blue) and a simulator without drift (grey). **c**, Parameters calibrated separately on early and late sessions.

Discussion

Earlier long-term studies each described one dataset [19,21,22]. We put three monkeys and six people on the same footing, with one decoder, one set of corrections and one inclusion rule, and added electrode statistics from 14 people. Seen this way, decay curves differed widely between individuals, and part of the decay reported for a decoder came from its regularization rather than from the recordings. Renormalization, not a fixed or mean-only decoder, is the baseline a label-free method has to beat. An alignment fitted with labels recovered far more than the label-free methods did, yet left part of the loss that no linear alignment of the dominant subspace recovered. Per-channel corrections need a scrambled-channel control before they are read as per-channel drift: in people, even scrambled decoders could be rescued. At the electrodes, a permutation null separated brief silences, which looked like noise, from month-long ones, which were real and often reversed.

For building and testing decoders, the results point to a few habits. Tune the ridge penalty on training-day data before reporting decay. Report label-free methods against renormalization. In people, recalibrate with about 100 labeled trials; at that budget a fresh, strongly regularized decoder did at least as well as a refit toward the old one, while the refit helped when only 10 to 20 trials were available. Set recalibration schedules per person, because one schedule will not suit individuals whose decoders lose half their accuracy within a day and others who keep it for months. Do not treat an electrode that falls silent for a few sessions as lost; most return. New methods are best compared against these steps.

Earlier work modeled per-electrode drift as shifts in baseline over weeks [15,24], and human decoders absorb day-to-day changes with input layers learned for each day [5,29]. Other studies describe changes in the preferred directions of individual features [35] and rotation of decoder weights [22]. In our data, decoders with correctly assigned channels beat scrambled ones, and re-weighting tracked each electrode's firing, so some of each channel's tuning pattern survived for months, most clearly in the monkeys. In the

monkeys, the best alignment within the population subspace also beat per-channel re-weighting, which points to activity being remixed across channels, as expected if population dynamics stay within a stable low-dimensional manifold[31,46]. In people the reverse held, which fits per-channel change, although a two-dimensional target limits this inference. For label-free methods, this means a better target exists: a method that found the alignment the labels reveal would recover much of the loss, but not all of it. Changes in the number or proximity of neurons near each electrode would produce such per-channel changes[47,48], but our data cannot identify the cause.

Activity near the yield threshold fluctuated strongly between sessions. Yield counts from single sessions are therefore noisy, and channels that fall silent for a few sessions are likely to return. Hahn and colleagues analyzed yield, decoding signal quality and tuning stability in the same human data[19]. We add a permutation null for silence and recovery and a link between electrode changes and decoder corrections.

All analyses were offline. Offline variability can overstate online decay[49], and in closed loop users adapt to their decoder[50]. Human decoding labels were inferred from the direction from cursor to target[23], and the online decoder shaped both the neural activity and the cursor path, so part of the cross-day change may reflect the user's control strategy or the task, not only the recordings. Three of the nine BrainGate participants with decoding data were excluded: in two the decoders did not work, and in T11 recordings jumped in scale between blocks, which session-wise normalization did not handle (Supplementary Note 4). Six participants and three monkeys cannot establish population rates, and the datasets differ in task, signal type, array placement and implant age. We did not test adversarial alignment. The foundation-decoder comparison was a pilot with one model size, threshold crossings and three individuals, and larger models or other ways of adapting them may do better. The simulator is calibrated to feature-level statistics in one monkey and does not model biophysics.

We release our code, derived results, the benchmark and the simulator, with the correction ladder as the baseline to beat. In these data, keeping implanted decoders working over months depended less on elaborate corrections than on a well-regularized decoder, renormalization of each session and brief recalibration with labeled trials.

Methods

Datasets and ethics

All data were public and de-identified. The original studies obtained ethical approval and informed consent, as described in the cited publications. We collected no new data.

Monkey N (LINK; DANDI Archive 001201[51]; ref.[21]). One rhesus macaque had Utah arrays in the hand area of primary motor cortex[38]. The release contains 312 sessions on 303 days over 1,242 days, in which the monkey moved two finger groups to targets in center-out or random-target blocks. We used the 96 released channels (64 on the medial array and 32 on the lateral array), with spike-band power and threshold crossings at -4.5 times the root-mean-square noise in 20 ms bins[52]. Outputs were the position and velocity of the two finger groups. Electrode impedance was stored on 228 days.

Monkeys C and M (DANDI Archive 000688[53]; refs.[31,37]). Both monkeys moved a cursor with a planar manipulandum in center-out and random-target reaching tasks. Monkey C had one Utah array in primary motor cortex (96 channels in these data), and monkey M had one in primary motor and one in dorsal premotor cortex. Monkey C had 68 sessions from October 2013 to October 2016, and monkey M had 28 sessions from January 2014 to June 2015. We summed the spikes of all sorted units on each electrode in 20 ms bins, giving 96 (C) or 192 (M) channels. Cursor position and velocity, which followed the hand, were interpolated to bin centers.

BrainGate participants (Dryad[54]; ref.[19]). The yield data contain one recording from each of 2,289 research sessions in 14 participants with 20 arrays (A1, S1–S3, T1–T3 and T5–T11). For every electrode, they list threshold-crossing rates at several thresholds (with the threshold in microvolts), impedance and grid position. Nineteen arrays were in the hand area of motor cortex and one was in the middle frontal gyrus. The decoding data contain closed-loop cursor sessions with spike-band power in 10 ms bins, cursor and target positions and the periods of attempted movement, for nine participants (T2, T3 and T5–T11). We used all sessions of each participant. Sessions, spans, arrays, trials per session and movement durations are listed in Supplementary Tables 1 and 14. All sessions of a participant were treated as one task type.

Inclusion of participants and session pairs. Retention is undefined when the reference decoder does not work. A participant therefore entered the decay and correction analyses only if the median R^2 of its reference decoder across session pairs exceeded 0.1. We fixed this rule when the first added participant (T3) failed, before the results of the other five added participants were seen. Six participants met it (T6, T5, T9, T7, T10 and T8). T2 and T3 did not (median reference R^2 −0.02 and −0.05). Their movement periods lasted a median of 15.6 s, against 1.6–3.9 s in the other participants. For T2, neither block-wise normalization nor threshold crossings recovered a usable decoder (Supplementary Table 14). T11 also failed the rule (median reference R^2 −16.19) because its recordings jumped in scale between recording blocks; it was analyzed with block-wise normalization as a sensitivity analysis (Supplementary Note 4). Within included individuals, session pairs whose reference R^2 was at most 0.1 were left out of retention summaries, in monkeys and humans alike. This rule was adopted after a single failed test session (reference R^2 −42.8) made bootstrap intervals meaningless; it removed 0–18% of pairs per individual and changed median retention by less than 0.05 in nearly all individuals and gaps. Results with all pairs are given in the Supplementary Information.

MINDFUL (Dryad[55]; ref.[35]). Closed-loop cursor blocks in which T11 (15 days) and T5 (6 days) used a fixed decoder, with neural features in 20 ms bins (spike-band power for T11 and threshold crossings for T5), the decoder's velocity output and the angular error of the cursor.

Features, decoders and accuracy

For monkey N and the human participants we used spike-band power (human 10 ms bins averaged in pairs). For monkeys C and M we used per-electrode spike counts. Each channel was z-scored with the mean and standard deviation of the decoding session's training segment, computed without labels (renormalization). The linear

decoder was ridge regression[56] on 8 causal lags (160 ms) of the z-scored features, with an unpenalized intercept. Linear decoders such as the Kalman filter are widely used for cursor control in people[57,58]. The default penalty was $\alpha = 0.1$ for monkey N and the humans, adopted from ref.[21], and $\alpha = 1$ for monkeys C and M; the tuned penalty was $\alpha = 10^4$ for the decoder trained on day i and for the reference decoder. Outputs were finger position and velocity for monkey N, cursor position and velocity for monkeys C and M, and for the humans the intended movement direction, defined as the unit vector from cursor to target during attempted movement, as in the calibration of human cursor decoders[23,57]. Accuracy was the coefficient of determination R^2, averaged over outputs and computed on held-out trials of the test session. In monkey N the first 300 of 375 trials trained the decoder, as in ref.[21]; in the other datasets the first 80% of trials did.

The recurrent decoder was a long short-term memory network[59] with 64 hidden units, input dropout[60] of 0.5 and 20-bin input windows (400 ms). It was trained with AdamW[61] (learning rate 10^{-3}, weight decay 0.01) for up to 4,000 iterations of 256 samples, with early stopping on the last 20% of training trials checked every 100 iterations. Without these measures, a 256-unit network memorized the human training sessions (training R^2 0.97, test R^2 -1.1 on one session). The network analyses used 25 training sessions per individual chosen at random with a fixed seed.

Session pairs

For each training session i and target gap $g \in \{1, 7, 30, 120, 480\}$ days, we selected the later session j of the same task type whose gap was closest to g, within 25% of g or one day, whichever was larger. The one-day set therefore contains one- and two-day gaps, which we report separately. In monkey N, 80 training sessions were drawn at random with a fixed seed; the full monkey N set (537 pairs) also includes further gaps and was used for monitoring. Pair counts are listed in Supplementary Table 2. The time to half retention was found by interpolating median retention at the target gaps linearly in log days, with 95% confidence intervals from a cluster bootstrap over training sessions. When median retention was already below one half at the one-day gap (pairs 1 or 2 days apart), we report less than one day; when it never fell below one half, we report the longest gap with data as a lower bound.

Correction ladder

Let $\mathbf{x}_j(t) \in \mathbb{R}^C$ be the raw features of test day j, $\boldsymbol{\mu}_d$ and $\boldsymbol{\sigma}_d$ the channel means and standard deviations of day d, and f_i the decoder trained on day i.

- *No correction*: $f_i\big((\mathbf{x}_j - \boldsymbol{\mu}_i)/\boldsymbol{\sigma}_i\big)$.
- *Mean updating*: $f_i\big((\mathbf{x}_j - \boldsymbol{\mu}_j)/\boldsymbol{\sigma}_i\big)$.
- *Renormalization*: $f_i(\mathbf{z}_j)$ with $\mathbf{z}_j = (\mathbf{x}_j - \boldsymbol{\mu}_j)/\boldsymbol{\sigma}_j$.
- *Re-weighting*: f_i applied to $g_c z_{j,c}$ for each channel c, with one offset per output. The weights g_c (unconstrained in sign unless stated) were shrunk toward 1 and the offsets toward the old intercept; the penalty, scaled by the number of samples, was chosen from $\{10^{-6}, \ldots, 10^{2}\}$ by cross-validation over five contiguous blocks of the first n trial indices. Sessions with fewer than n trials therefore had fewer, uneven

folds; when all their trials fell in one block, the blocks were formed from the available trials. Here and for the ridge-to-prior refit, the candidates also included an infinite penalty, which keeps the previous decoder, so that cross-validation could decline an update.

- *Ridge to prior*: a new decoder $(\mathbf{W}, \mathbf{b})$ fitted on the lagged features $\mathbf{H}$ of day j by minimizing

$$\|\mathbf{Y} - \mathbf{H}\mathbf{W} - \mathbf{1}\mathbf{b}^\top\|_F^2 + \lambda \left(\|\mathbf{W} - \mathbf{W}_i\|_F^2 + \|\mathbf{b} - \mathbf{b}_i\|_2^2\right), \tag{1}$$

  where $(\mathbf{W}_i, \mathbf{b}_i)$ is the old decoder and λ was chosen from $\{0.1, 1, \ldots, 10^6\}$ by cross-validation.
- *Retrained decoder*: ridge regression on day j's first n labeled trials, with α chosen from the same grid by cross-validation.
- *Reference*: a decoder trained on all of day j's training trials.

Supervised rungs used the first n labeled trials of day j ($n = 300$, or all training trials if fewer, unless stated). Retention was the ratio of a rung's R^2 to the reference R^2 on the same test trials, and the share of the loss recovered by re-weighting was $(R^2_{\text{reweight}} - R^2_{\text{renorm}})/(R^2_{\text{ref}} - R^2_{\text{renorm}})$. For the network decoder, re-weighting was a learned gain and offset per channel in front of the frozen network, and the refit was fine-tuning with an L_2 penalty toward the old weights; penalties were chosen on the last 20% of labeled trials. We also computed a full input remap fitted by L-BFGS[62]; because it can represent any linear decoder of the same form, we used it only as a calibration target for the simulator.

Channel controls. For 40 training sessions per individual (22 for monkey M and 33 for T7, all they had; fixed seed), re-weighting was repeated with weights constrained to be non-negative (bounded least squares with the cross-validated penalty), and for a copy of the day-i decoder whose channel weights were randomly permuted, with signed and with non-negative weights.

Electrode link. For the same pairs in the human participants, the signed re-weighting weights were correlated (Spearman) across channels with the change in log threshold-crossing rate ($\log(\text{rate} + 0.1)$ at -4.5 times the noise) and in log impedance between the yield recordings of the two days, matched by electrode identity. A null correlation was computed after permuting channels.

Matched-output control. For monkeys C and M, sessions were rebuilt with a two-dimensional unit-vector target, the direction of cursor velocity, on bins where cursor speed exceeded the session's 25th percentile, and the ladder was run on the same pairs with the tuned penalty.

Decoder-weight angles. Each channel's block of ridge weights (8 lags by all outputs) was stacked into $\mathbf{B}_i$ and $\mathbf{B}_j$. We computed the angle between $\text{vec}(\mathbf{B}_i)$ and $\text{vec}(\mathbf{B}_j)$, which allows one global scale, and the angle between $\text{vec}(\mathbf{B}_j)$ and $\text{vec}(\text{diag}(\mathbf{s})\mathbf{B}_i)$, with each channel's scale s_c fitted by least squares.

Example in Fig. 1. Waveforms are the released robust mean waveforms (15 kHz, in microvolts) for electrodes with a threshold-crossing rate of at least 2 Hz at -4.5 times the robust standard deviation. Amplitude is the peak-to-peak value of these waveforms. Rates in panel d are the released threshold-crossing rates at the same threshold,

averaged within 30-day bins. The cortical surface was rendered with PyVista, with the precentral gyrus taken from the Desikan–Killiany atlas. Array sites were placed on its crown near the hand knob by MNI coordinates and are approximate. The array inset is a rendering from published dimensions, not a photograph. Silhouettes are from PhyloPic (CC0). For the reconstructions, decoders used the tuned penalty: one trained on T5's day 355, one trained on day 833, and the day-355 decoder re-weighted with 300 labeled trials from day 833. Paths are shown for held-out movements that started at the center. Each path integrates the decoded direction from the cursor's start position. The step size was set so that the intended direction would reach the target in the trial's duration, multiplied by a common display gain of 2 for all decoders. R^2 values are those of the analysis on the same held-out trials.

Label-free corrections

Procrustes rotation. Let $\mathbf{V}_i, \mathbf{V}_j \in \mathbb{R}^{C\times k}$ hold the top $k = 16$ principal axes of the corrected training features on each day. The orthogonal matrix $\mathbf{R}$ minimizing $\|\mathbf{V}_j\mathbf{R} - \mathbf{V}_i\|_F$ was found by Procrustes analysis[40], and inputs were mapped as $(\mathbf{I}-\mathbf{V}_j\mathbf{V}_j^\top)\mathbf{z} + \mathbf{V}_i\mathbf{R}^\top\mathbf{V}_j^\top\mathbf{z}$. We applied it to renormalized features and, separately, to mean-updated features with $\mathbf{V}_j$ estimated from those features.

CORAL. Renormalized day-j features were re-colored to match day i's covariance, $\mathbf{z}' = \mathbf{C}_i^{1/2}\mathbf{C}_j^{-1/2}\mathbf{z}$[41], with covariances shrunk 5% toward a scaled identity.

Factor-analysis stabilizer. Following ref.[30], a factor-analysis model with 10 latent factors was fitted to each day's renormalized training features, and a ridge decoder on 8 lags of the day-i posterior latent means was trained. On day j, a model fitted without labels was aligned to day i by orthogonal Procrustes analysis of the loadings of stable channels, found by iteratively keeping the 60% of channels with the smallest loading residuals; latents were then inferred from stable channels only and passed to the frozen latent decoder. The comparison baseline reused the day-i factor model on day j.

All label-free comparisons were summarized per training session over its pairs at all gaps.

Alignment ceiling. To estimate how much any linear alignment of the dominant subspace can recover, we fitted the best such alignment with labels. Inputs were mapped as $(\mathbf{I} - \mathbf{V}_j\mathbf{V}_j^\top)\mathbf{z} + \mathbf{V}_i\mathbf{Q}\mathbf{V}_j^\top\mathbf{z}$ in front of the unchanged day-i decoder, as for the Procrustes rotation, but with $\mathbf{Q}$ a free 16×16 matrix fitted to day j's labeled trials, which allows rotation, scaling and mixing within the subspace. $\mathbf{Q}$ was shrunk toward the Procrustes solution and the offsets toward the old intercept, with the penalty chosen as for re-weighting. It was fitted on all of day j's labeled training trials, the same trials the reference decoder uses, and re-weighting was refitted on the same trials for comparison. Both were scored on the held-out test trials, so the ceiling is an estimate for linear alignments of this subspace, not a strict bound for every method. For gaps of 120 and 480 days, the share of the loss recovered by a correction was computed from medians over pairs, $(\tilde{R}^2 - \tilde{R}^2_{\text{renorm}})/(\tilde{R}^2_{\text{ref}} - \tilde{R}^2_{\text{renorm}})$, with 95% intervals from a cluster bootstrap over training sessions. Re-weighting and the ceiling were compared per training session over all gaps (two-sided Wilcoxon signed-rank

test, Holm correction across individuals). The label-free columns and the reference decoder reproduced the main analysis exactly.

Monitoring analyses

In monkey N, for every pair we computed Gaussian Kullback–Leibler divergences between days i and j of the projections of renormalized features on day i's top ten principal axes (neural divergence) and of the decoder's outputs (output divergence). We report raw Spearman correlations with R^2, partial correlations given log elapsed days and time-blocked cross-validated linear prediction of R^2. The pilot statistics (first 46–61 sessions, 283 pairs) are described in Supplementary Note 1. For the MINDFUL data we followed the original definitions[35]: non-overlapping 60 s windows, divergences against the first day, and the median angular error over non-excluded trials (windows with fewer than 500 valid bins skipped). We report Pearson and partial correlations given elapsed days across windows, and leave-one-day-out linear prediction of angular error; the reference day was excluded from evaluation.

Regularization and recalibration

Ridge decoders were trained with $\alpha \in \{0.1, 10, 10^3, 10^4, 3 \times 10^4, 10^5, 3 \times 10^5, 10^6\}$ on 40 training sessions per individual (22 for monkey M and 33 for T7) and evaluated on held-out same-day trials and on later sessions at the five gaps. The 1% rule chose the largest α whose median same-day R^2 was at least 99% of the best median. These analyses compare absolute R^2 between penalties rather than retention, so the pair rule was not applied. Compared with each training session's own best same-day penalty, the rule changed cross-day R^2 by a median of 0.004 in monkey N ($P = 6.9 \times 10^{-5}$) and 0.000–0.013 in the participants, and not at all in monkeys C and M. For the human data-efficiency analysis, the ladder was run for 30 training sessions per participant (fixed seed) with $n \in \{10, 20, 50, 100, 300\}$. So that every budget used the same pairs, we kept only pairs whose later day had at least 300 labeled training trials (245 of 467 valid pairs); sessions in T6, T9 and T7 are often shorter, so these participants contribute fewer pairs. For harmful recalibrations we used 86 monkey N pairs with default-penalty decoders; the shrinkage λ in equation (1) was chosen per pair by cross-validation on the n trials, or as the value with the best median R^2 over pairs from other training sessions, which was 10^4 for every pair with up to 50 trials. A recalibration was harmful when its R^2 was more than 0.01 below that of the renormalized fixed decoder.

Electrode statistics

Statistics were computed per array. An electrode was active in a session when its threshold-crossing rate at -4.5 times the root-mean-square noise was at least 2 Hz. An initially active electrode (mean rate of at least 2 Hz over the first three sessions) was silenced at its first run of at least three consecutive observed sessions below threshold, and recovered when it later exceeded 4 Hz for at least three consecutive sessions. Missing sessions were skipped, not imputed. Control analyses (Supplementary Table 11) varied these definitions. They used thresholds of -3.5 and -5.5 times the noise, and a fixed threshold per electrode in microvolts: its -4.5 value over the first three sessions,

applied by choosing in each session the noise multiple whose microvolt threshold was closest. They also removed sessions with array-wide shifts (absolute median change in log rate above 1), required a mean waveform trough of at least 30 μV, required silence below 0.5 Hz, or required silences lasting at least 30 days. The null distribution permuted each electrode's session order independently, 200 times; we report the observed recovery fraction, the null mean and 95% range, and a one-sided permutation P value. For impedance, values of 4,000 kΩ or more were excluded and the trend was the Spearman correlation of the array median with day. For the edge effect, edge electrodes were those on the outer ring of the grid (10×10 in humans and 8×8 in monkey N), each initially active electrode's decline was the slope of $\log(\text{rate} + 0.1)$ against years, and arrays with at least 10 initially active electrodes were tested with the array as the unit. The direction was predicted from mechanical models[45], so we report one-sided tests; two-sided P values are twice as large. The adjusted analysis regressed slopes on edge membership and initial log rate within each array.

Simulator

The simulator transforms the z-scored features $\mathbf{Z} \in \mathbb{R}^{T \times C}$ of a real base session into those of a simulated session Δt days later (Supplementary Note 2). *Mixing*: $\mathbf{Z} \leftarrow \mathbf{Z}(\mathbf{I} + \mathbf{E})^{\top}$, with $\mathbf{E} = s(\Delta t)[(1-\beta)\mathbf{G} + \beta\mathbf{L}]$, where $\mathbf{G}$ is a Gaussian matrix with zero diagonal, $\mathbf{L}$ a Gaussian matrix restricted to grid neighbors, $\beta = 0.5$ and $s(\Delta t)^2 = s_0^2 + [s_\infty(1 - e^{-\Delta t/\tau_s})]^2$. *Turnover*: each channel replaces a fraction ρ_c of its signal by a unit with rotated tuning in the session's 20-dimensional latent space; ρ_c follows a beta distribution with mean $\rho_0 + (\rho_\infty - \rho_0)(1 - e^{-\Delta t/\tau_\rho})$ and concentration 5. *Silencing*: each channel active in the base session becomes silent with probability $h_{\text{off}}/(h_{\text{off}} + h_{\text{on}})\,[1 - e^{-(h_{\text{off}}+h_{\text{on}})\Delta t}]$ and keeps 37% of its signal variance, the ratio of tuning strength of silent to active channels in monkey N. The switching rates ($h_{\text{off}} = 0.0038$ and $h_{\text{on}} = 0.0018$ per day) were measured from monkey N's activity in the calibration period. The six mixing and turnover parameters were calibrated on 12 base sessions before day 700 by minimizing the summed squared difference between simulated and real median retention for three rungs (renormalized, re-weighted and full input remap) at five gaps, with the same correction code and regularization as for real data, using a 24-point Sobol search[63] followed by Nelder–Mead[64]. Validation used 10 base sessions after day 700 and compared predicted and real medians for all rungs and for $n \in \{20, 50, 100\}$; rungs not fitted were the label-free alignments, the subspace rotation fitted with labels and the two refits. The empirical baseline predicted each late-period median by the corresponding median over pairs whose training session preceded day 700. Augmentation added eight simulated futures per training session (Δt log-uniform between 1 and 120 days) and was compared with stronger regularization, random channel dropout with gain noise[26] and training on the three previous sessions.

Pretrained foundation decoder

We tested NDT3 (ref.[43]), a transformer pretrained on about 2,000 hours of intracortical motor recordings, using its public 45-million-parameter checkpoint (code at

commit 877f9f5, run with the authors' software versions). The checkpoint stores the list of its pretraining sessions. It includes the datasets of monkeys C and M, but no BrainGate or monkey N sessions, so we tested monkey N and two BrainGate participants. NDT3 takes threshold-crossing counts, so all comparisons in this section used threshold crossings (20 ms bins; -4.5 times the RMS for BrainGate), with the ridge decoder of the correction ladder fitted to the same features. We chose participants whose threshold crossings decoded well (T5 and T8; same-day R^2 of the ridge decoder 0.50 and 0.32 over all their sessions, against 0.45 and 0.28 on the sessions of the pilot) and 12 session pairs per individual: the pairs of three training sessions spread over the recordings, each with pairs at four or five of the main gaps.

Sessions entered NDT3 through its own data pipeline (1-s windows, its tokenization of 32-channel patches, its training step and prediction function), so that the model saw data as in its original evaluations. Behaviour was scaled by the 99.9th percentile of its magnitude on the training day only, and R^2 was computed on unscaled labels. Two rules were fixed before the session pairs were run. The learning rate was chosen per individual from 10^{-4} and 4×10^{-4} by NDT3's own validation R^2 on its earliest session (4×10^{-4} for monkey N and T5, 10^{-4} for T8). An array was left out of NDT3's input when fewer than 25% of its channels were active on that session. We wrote this rule after a diagnostic showed that NDT3 failed to train on T8 while one nearly silent array (17 of 96 active channels) was present (Supplementary Note 5). It removed only that array. The ridge decoder always used all channels.

For each training session, the pretrained model was fine-tuned on day i (all labeled trials; early stopping on a random 20% of 1-s windows). On day j it was applied unchanged, fine-tuned further with the first 50 or 300 labeled trials, or adapted without labels by continuing training on day j with its neural prediction loss only. A model fine-tuned on day j alone gave the same-day reference for retention. Predictions were causal: overlapping 2-s windows advancing by 1 s, keeping the last second of each. This gave R^2 within 0.01 of NDT3's streaming evaluation (Supplementary Note 5).

Benchmark

The benchmark fixes the session pairs of the tuned-penalty analysis at the five main gaps: one pair per training session and gap, with the closest later session of the same task within 25% of the gap. For each pair, a method may use the training segment of day i with its kinematics, the training segment of day j without kinematics and the kinematics of its first n trials ($n = 0$, 10, 20, 50, 100 or 300), and the neural features of day j's evaluation segment. It returns predictions for that segment, which are scored by R^2 and by retention relative to a same-day ridge decoder (8 lags, penalty 10^4). Pairs count when that reference exceeds $R^2 = 0.1$. Summaries report medians with cluster bootstrap intervals over training sessions. The reference methods are the rungs of the correction ladder. Through the benchmark's interface they reproduce the values of this study to within 10^{-14}. A threshold-crossing version of the features is included for decoders that require spike counts.

Statistics

Session pairs that share a training session are not independent. Confidence intervals for medians therefore came from a cluster bootstrap over training sessions[65] (2,000 resamples, percentile intervals), and paired tests used one value per training session (two-sided Wilcoxon signed-rank tests). Pairs also share test sessions, so these intervals and P values may be somewhat optimistic; we emphasize effect sizes. Harmful recalibrations were compared per training session (the share of its pairs made worse by each policy; two-sided Wilcoxon signed-rank test over training sessions), Holm-adjusted over the five data budgets[66]. Counts of significant individuals are also given after Holm correction across individuals. With six participants and three monkeys, comparisons between species are descriptive. Exact P values are reported. Analysis choices that changed during the study, and why, are listed in Supplementary Note 3.

Whether decay differed between individuals was tested with a linear mixed model on the valid pairs of the better-regularized decoder ($n = 1,684$ pairs). The outcome was the retention of the renormalized fixed decoder, floored at 0 so that every failed decoder counts as no retention (the raw ratio reaches -50 when a broken decoder is divided by a small reference). The predictor was $\log_2$ of the days between sessions. The full model had an intercept and slope per individual; random intercepts for the training session and for the test session were crossed, because pairs share both. Models were fitted by maximum likelihood and compared by likelihood-ratio tests against one shared curve (curves differ) and against individual intercepts with one shared slope (slopes differ). Sensitivity analyses used a floor of -1 and the absolute loss in R^2 (reference minus renormalized decoder), which needs no ratio. Analyses used Python 3.11 with NumPy, SciPy, pandas, scikit-learn, statsmodels and PyTorch.

Use of AI tools

An AI assistant (Claude, Anthropic) was used to write analysis code, run analyses and draft the text under the authors' direction. The authors reviewed the code, analyses and text and take full responsibility for the content.

Data availability

All data analyzed in this study are public. LINK (monkey N) is available from the DANDI Archive at https://doi.org/10.48324/dandi.001201/0.251023.2336. The recordings of monkeys C and M are available at https://doi.org/10.48324/dandi.000688/0.250122.1735. The BrainGate yield and decoding data are available from Dryad at https://doi.org/10.5061/dryad.x0k6djj1h. The MINDFUL data are available from Dryad at https://doi.org/10.5061/dryad.n2z34tn5s. All derived results underlying the figures and tables are provided in the code repository.

Code availability

All code, including the benchmark (fixed splits, data interface, reference methods and scoring) and the simulator with its calibrated parameters, is available at https:

//github.com/Imtiaj-Sajin/invasive-bci under the MIT license. A versioned archive will be deposited on Zenodo upon publication.

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Competing interests. The authors declare no competing interests.

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